Vector Databases

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Presentation outline

1: What is a vector database

2: Why is the technology relevant?

3: How does it work

4: What are some alternatives?

5: What is a use cases for a Vector database?

6: Demo of code

7: Quiz: Introduction to Vector Databases

1: What is a vector database?

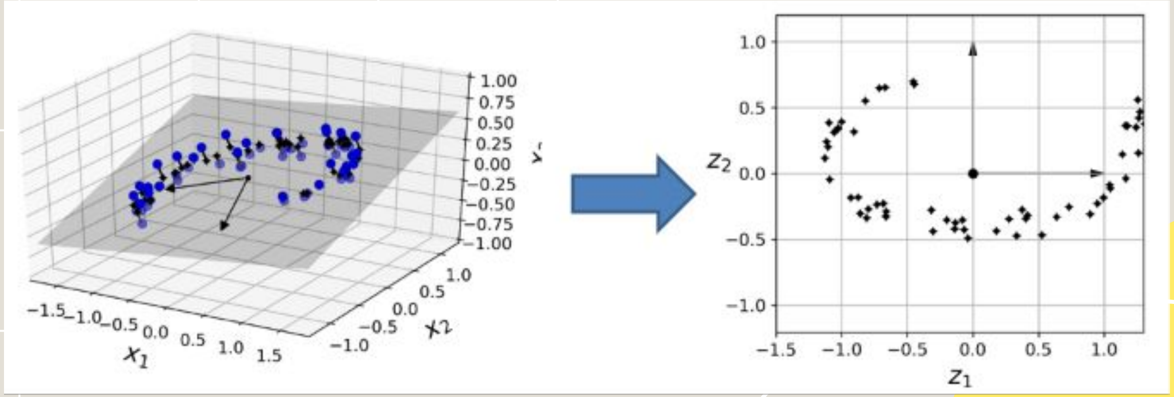
Vector: quantity with both **magnitude(size & length)** and direction

Words have magnitude and direction:

“it’s not bad”	->	<-	“it’s not good”
“i like it”	—→	←-	“I dislike it”
“i love it”	------>	←—---	“I hate it”
“I’m obsessed”	—--------->	←—-------	“i despise it”

Example of high dimensional data:

“I dislike dogs” VS “i love cats” VS “fish is tasty”



Source: Douglas College, 2025

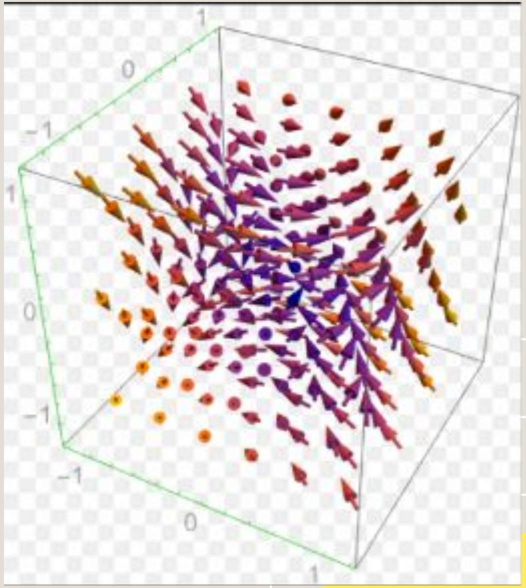
2: Why is the technology relevant?

Data Is enriched by being compared in higher dimension.

- **Semantic search and similarity matching:** find content based on meaning and not key words.
    - **Example:** “House price”**<- “**I like animals **” ——-->** **“**i like cats”

- **Nearest neighbour search for recommendation systems :** find similar data points to cluster.
    - **Example:** “i like lons” ←–– **“**i like felions“ —----> “I like cats”

- **Retrieval Augmented Generation (RAG):** Make the LLM provide answers based on relevant embedded/vectorized documents.
    - **Example:** “find a document on felions” —> documents on cats/lions/panthers/….



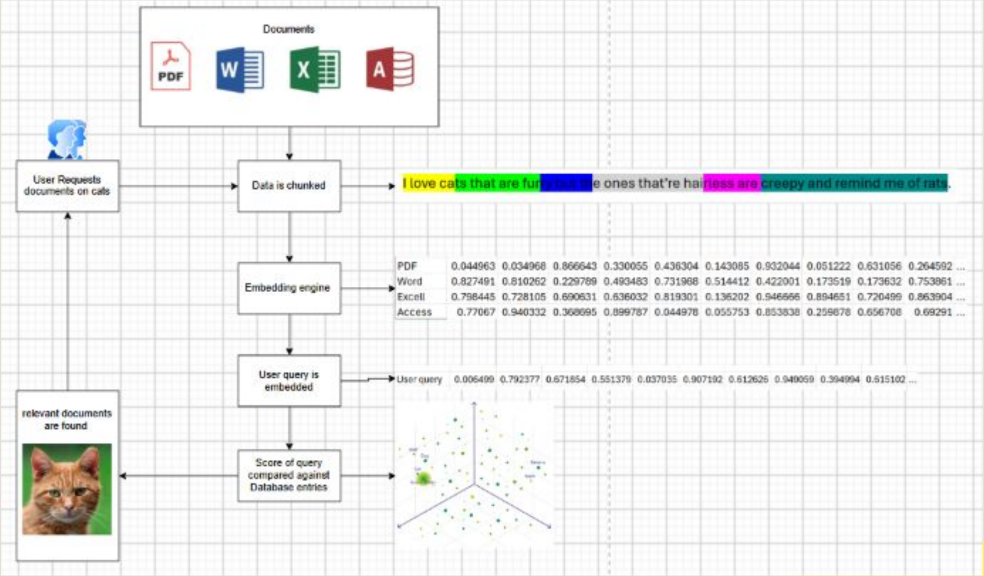
Source: *Wolfram 2014*

3: How does it work?

Chunk overlap: semantics are captured across clusters of words.

Embeddings: Semantic encoders extract meaning from words.

Scoring: Calculate the relevance score of the query vs existing documents.



Source: Cristi S, 2026

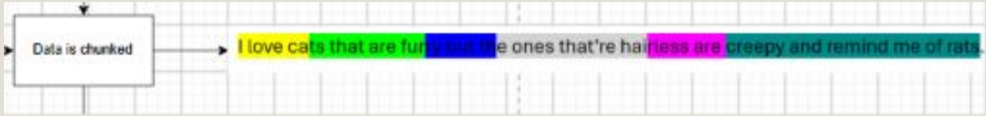
3A: Chunk overlap

100 page document is chunked into a page.

Each page is further chunked into paragraphs, then sentences.

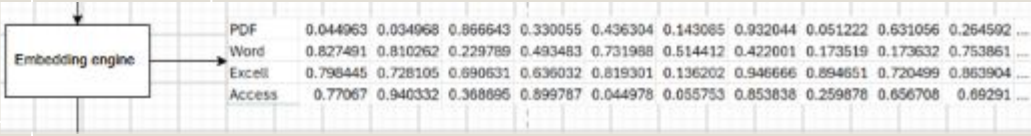
Each sentences is chunked into parts based on meaning and intensity.

Chunks overlap to retain inter-chunk context.



Source: Cristi S, 2026

3B: Embeddings



PDF	0.044963	0.034968	0.866643	0.330055	0.436304	0.143085	0.932044	0.051222	0.631056	0.264592	...
Word	0.827491	0.810262	0.229789	0.493483	0.731988	0.514412	0.422001	0.173519	0.173632	0.753861	...
Excell	0.798445	0.728105	0.690631	0.636032	0.819301	0.136202	0.946666	0.894651	0.720499	0.863904	...
Access	0.77067	0.940332	0.368895	0.899787	0.044978	0.055753	0.853838	0.259878	0.656708	0.69291	...

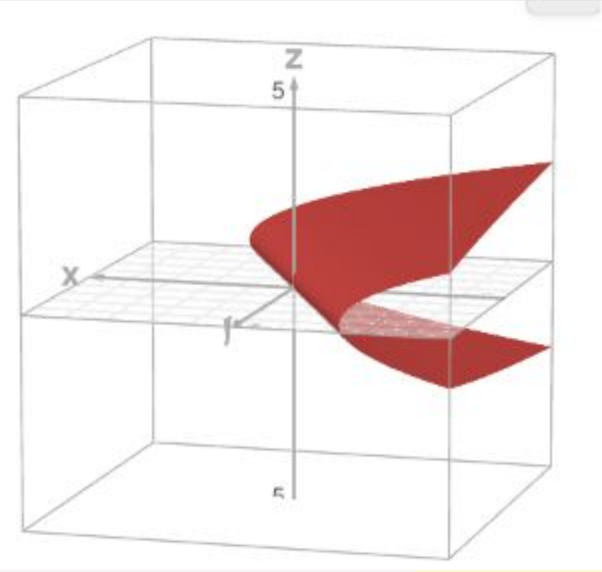
Source: Cristi, 2026

Transformers process chunks into vectors using self-attention mechanism for context.

- **Example**: “Bank account” vs “river bank”

Neural Networks like BERT can be used to create embeddings.

Vectors are scatters on the 3d coordinate grid



Source: Desmos, 2026

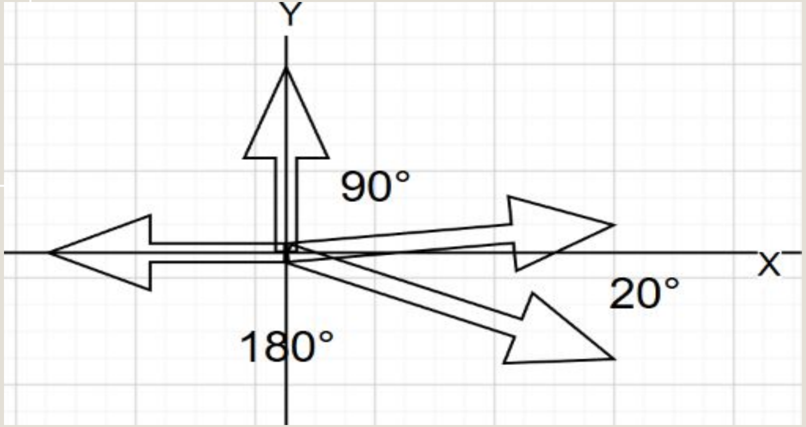
3B: Scoring

Storage and Retrieval is based on finding nearest neighbour vectors that share vector direction.

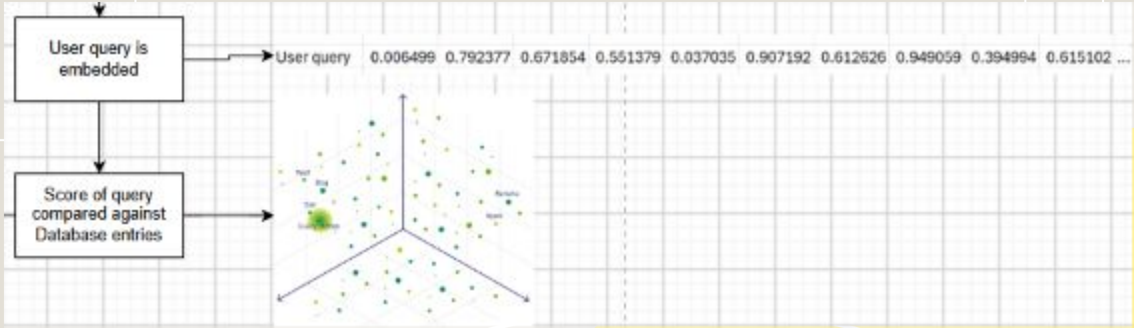
A query is embedded and its vectors are scattered, finding the nearest neighbours.

Vectors can capture momentum of a topic based on length of vector.

- **Example:** 15, 25, 150 users asking for cat images.



20° High similarity
90° No similarity
180° Perfectly opposite



Source: Cristi, 2026

4: What are some alternatives?

ChromaDB: Open source database that uses LangChain and LlamaIndex to support virtualization.

- **Key feature:** Ease of use.

Pinecone: close source that excels at high-dimensional data. Excels at machine learning applications.

- **Key feature :** performance at scale.

Weaviate: Open source AI native vector database, which allows the storage of data objects and vector embeddings. Excels at searching vectors.

- **Key feature:** store objects with metadata, properties, and relationships for complex data rich search.

Source: Chroma , 2026

Source: Pinecone, 2026

Source: Weaviate, 2026

5: Use case for vector databases.

Problem

Traditional databases and keyword-based search systems retrieve information mainly through exact term matching. This approach may fail when a query and a relevant document express the same concept using different vocabulary.

For example:

Query: **“How can I reduce cloud infrastructure costs?”**

Document: **“Strategies for optimizing computing resource expenditure”**

Although semantically related, the texts share few exact keywords.

Traditional keyword search may fail to retrieve all relevant listings because it depends on lexical overlap rather than semantic similarity.

By converting each property listing into a vector **embedding**, the system performs **semantic search**, retrieving results based on meaning rather than exact keywords.

5: Use case for vector databases.

Problem

Organizations generate thousands of unstructured marketing assets, including:

- Campaign briefs
- Product descriptions
- Email campaigns
- Presentations
- Brand guidelines
- Social media content

Traditional keyword search may fail when relevant materials express the same idea using different terminology.

By converting each file or text into a vector **embedding**, the system performs **semantic search**, retrieving results based on meaning rather than exact keywords.

5: Use case vector databases.

example

If there’s a query "Sustainability campaign" .

Documents may instead contain:

- Green initiatives
- Eco-friendly products
- Environmental responsibility
- Carbon reduction strategy

Although semantically related, exact keyword search may miss relevant documents.

5: Use case for vector databases.

Marketing Content Pipeline

Embedding Representation

$x_i \to e_i \in R^d$

where

- xix_ixi = document chunk
- eie_iei = embedding vector
- ddd = embedding dimension

Each document is transformed into a numerical representation that captures semantic meaning.

5: Use case for vector databases.

Semantic Retrieval

The user query is encoded using the same embedding model:

$q \rightarrow eq \in Rd$

The vector database compares the query embedding against all indexed vectors using similarity metrics such as:

- Cosine Similarity
- Euclidean Distance
- Dot Product

Approximate Nearest Neighbour (ANN) indexing enables efficient similarity search over millions of vectors.

Retrieval Function

$R(q)=TopKei \in D[sim(eq,ei)]$

The system returns the **Top-K** most semantically relevant documents.

5: Use case for vector databases.

Business Benefits

- Faster document discovery
- Reduced duplicated work
- Knowledge reuse
- Context-aware search
- Better decision support
- Improved productivity

Example Query

"Find previous sustainability campaigns targeting young consumers."

Returned Results

✓ Eco-Friendly Summer Campaign

✓ Green Product Launch

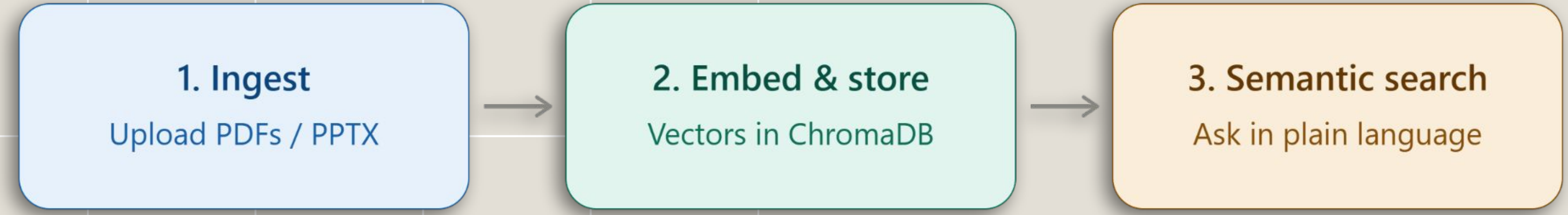
✓ ESG Brand Strategy

5: Use case for vector databases.

Real-World Applications

- Marketing asset management
- Enterprise knowledge bases
- Customer support systems
- Retrieval-Augmented Generation (RAG)
- Document recommendation systems
- Intelligent enterprise search

6: Demo code-ChromaDB Vector Search



Semantic search — find by meaning, not by keyword

What we're demonstrating:

1. Ingest Documents — Upload course PDFs / PPTX into the UI

2. Embed & Store — Convert text into vector embeddings in ChromaDB

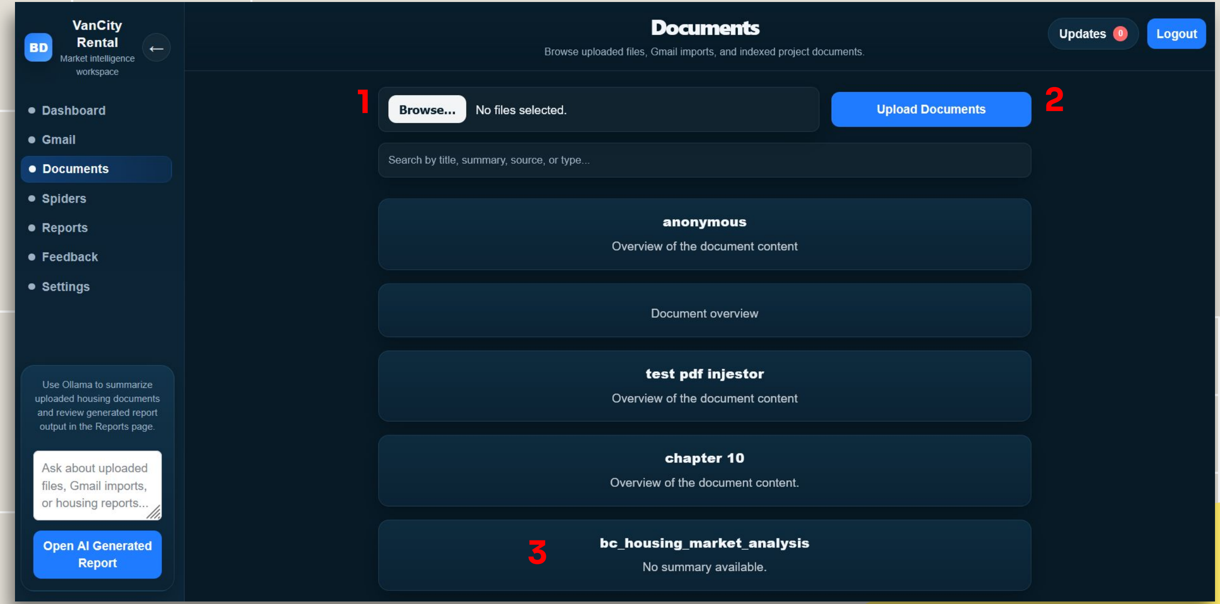
3. Semantic Search — Ask in natural language, retrieve by meaning Key idea: Keyword search finds words. Vector search finds meaning.

6: Step 1: Ingest Documents

1. Upload course PDFs / PPTX into the UI

2. Extract text from the files (OCR for scanned documents)

3. Documents enter the pipeline, ready for embedding

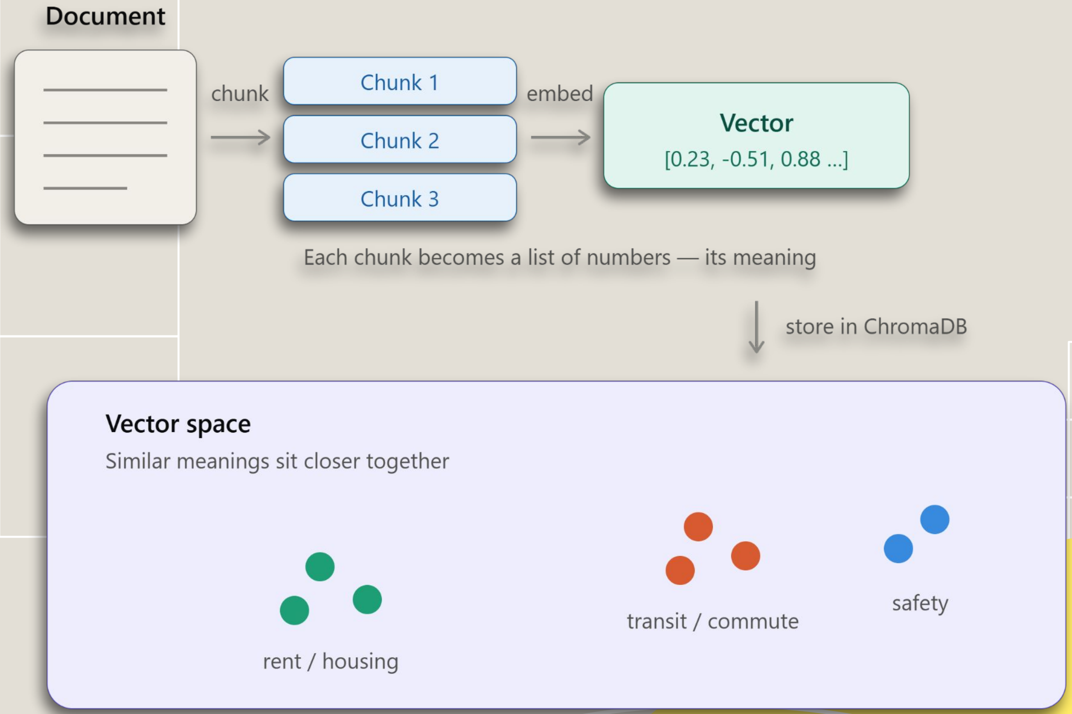


6: Step 2 — Embed & Store

1. **Chunking: split long documents into meaningful segments**

2. **Each chunk is converted into a vector embedding (via Ollama)**

3. **Vectors are stored in ChromaDB — similar meanings sit closer together**

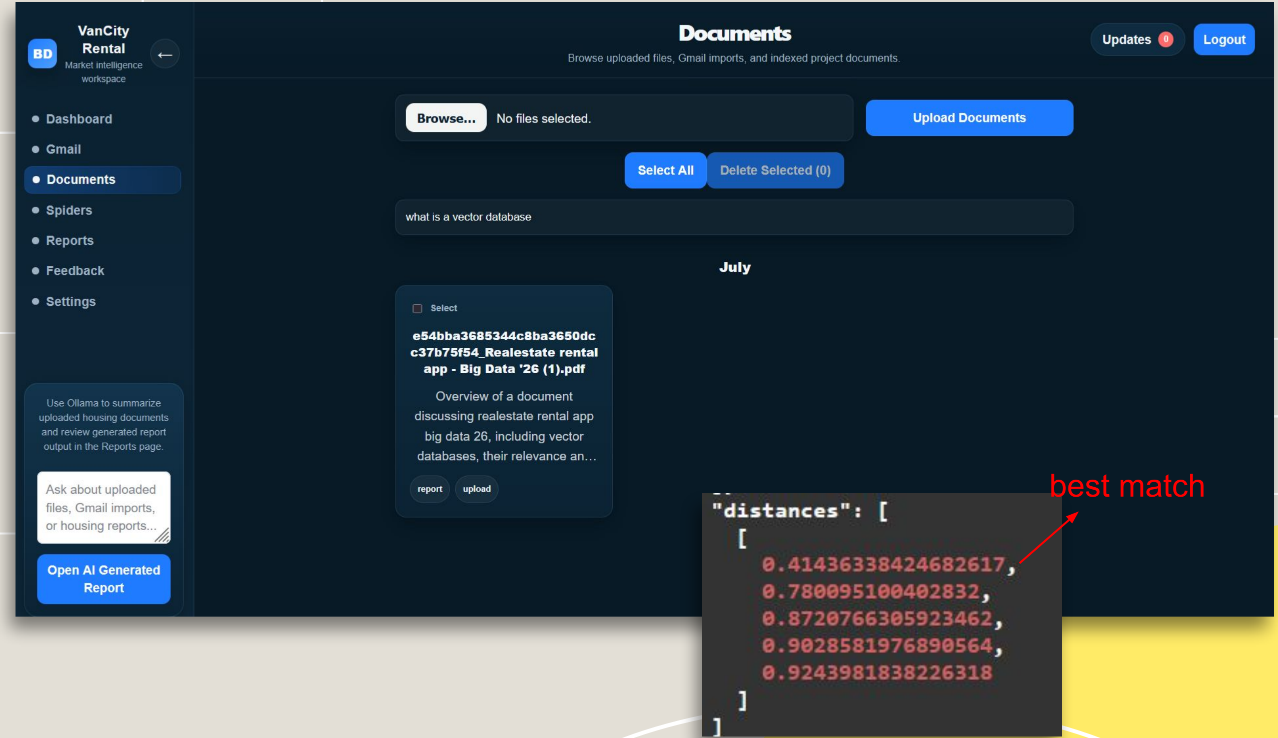


6: Step 3 — Semantic Search

1. Ask in natural language — no exact keywords needed

2. The query becomes a vector; ChromaDB finds the closest matches

3. Distance score = semantic relevance (smaller = more relevant)



7: Quiz: Introduction to Vector Databases

Ready? Scan now and let's start!

2. Choose Your Format

Groups can choose one of the following three tracks for their seminar:

- **The Research Deep-Dive:** Analyze a recent peer-reviewed paper (published within the last 24 months) regarding Big Data architectures, AI integration, or data ethics.
- **The Technology Spotlight:** Introduce a new or emerging tool/framework (e.g., DuckDB, Polars, dbt, or Vector Databases) and explain its position in the ecosystem.
- **The "Flash Course" (Hands-on):** A practical, tutorial-style session teaching the class how to use a specific library or tool to solve a data problem.

3. Mandatory Content Requirements

- The "Why" Factor: Why is this research/technology relevant right now?
- Technical Architecture: A clear explanation of how the system/concept works internally.
- Comparative Analysis: How does this compare to existing solutions (e.g., Spark vs. Flink)?
- Interactive Element: You must include a Q&A, a live poll, a short quiz, or a code demo to engage the audience.

4. Deliverables

- Presentation Slides (PDF or PowerPoint).
- Seminar Summary (1-page handout for classmates).
- Resource List (links to documentation, papers, or Github repos).

Thank you

Questions?